Complex Engineering Problem (CEP) Report

Course: CEE 417 - Geometric Design of Highway and Street

Instructor: Dr. Danish Farooq

Project Title: Investigate the critical geometric design features of Taxila GT Road and assess their impact on road safety

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1. EXECUTIVE SUMMARY

This report presents a comprehensive geometric design analysis of a selected segment of the Taxila GT Road, undertaken as part of Complex Engineering Problem (CEP) requirements for CEE 417 – Geometric Design of Highways and Streets, under the instruction of Dr. Danish Farooq. The study integrates field-surveyed geospatial data, AutoCAD Civil 3D digital modeling, and AASHTO Green Book standards to systematically evaluate the existing road geometry and propose evidence-based safety improvements.

Utilizing coordinate points imported directly from Google Earth, a Triangulated Irregular Network (TIN) Digital Terrain Model (DTM) was constructed, enabling the extraction of horizontal alignment, vertical profile, and transverse cross-sections. The horizontal alignment, exported as an alignment grid view (Civil 3D Worksheet), comprises 23 entities — 12 tangent lines and 11 horizontal curves — spanning a total centerline length of approximately 1,543.59 ft across stationing 0+00.00 ft to 11+63.39 ft. The design speed adopted for this study is 37 mph (≈ 60 km/h), consistent with semi-urban arterial highway classification.

Geometric analysis against AASHTO standards reveals that all 11 horizontal curves possess a consistent radius of 200.00 ft, except C1, C4, and C5 at 117.71 ft. At the adopted design speed of 37 mph, the AASHTO-required minimum radius for emax = 6% is 459 ft, which indicates a critical geometric deficiency across all curves in the alignment. The vertical profile reveals significant elevation variation between approximately 1,560 ft and 1,650 ft over the study corridor, presenting gradient challenges for heavy vehicle operations. Detailed cross-sections at stations 0+30.00 and 0+60.00 illustrate how the proposed lane geometry interacts with the existing ground surface.

All referenced Civil 3D drawings — including the road alignment and offset plan, contouring plan, surface profile view, transverse cross-sections, imported Google Earth points, and corridor model — are appended at the end of this report as supporting technical exhibits.

2. INTRODUCTION AND PROJECT OBJECTIVES

2.1 Background and Significance

The Taxila GT Road is a critical arterial highway corridor in the Rawalpindi-Taxila region of Punjab, Pakistan. As a principal inter-city connector, it accommodates a heterogeneous traffic stream comprising passenger vehicles, motorcycles, light commercial vehicles, and heavy goods trucks. The road's geometric configuration —

shaped partly by historical alignment decisions and constrained topography — presents measurable operational and safety challenges that warrant rigorous engineering scrutiny.

Geometric design constitutes the spatial framework of a highway: it defines sight distances, speed consistency, vehicle stability on curves and grades, and driver workload. When geometric design elements fall short of prescribed standards, the probability of run-off-road accidents, rear-end collisions, and vehicle overturning events increases substantially. The AASHTO Green Book (A Policy on Geometric Design of Highways and Streets) provides the internationally recognized benchmark for evaluating and correcting such deficiencies.

2.2 Project Objectives

The primary objectives of this Complex Engineering Problem (CEP) study are:

1. To construct a Digital Terrain Model (DTM) of the selected Taxila GT Road segment using coordinate data imported from Google Earth into AutoCAD Civil 3D.
2. To develop a full horizontal alignment and vertical profile of the existing road geometry and export the alignment data for quantitative analysis.
3. To compare the existing geometric design parameters — curve radii, superelevation, gradients, and sight distances — against AASHTO minimum standards for the applicable design speed and terrain classification.
4. To identify critical geometric deficiencies and assess their implications for road safety.
5. To propose AASHTO-compliant design improvements to rectify identified deficiencies.

2.3 Scope of Work

The study scope encompasses a road segment of approximately 1,543.59 ft (≈ 470 m), spanning from Station 0+00.00 to 11+63.39 ft. The terrain is characterized as semi-urban with plain-to-rolling topography. The study does not include intersection design, drainage design, or pavement structural design; it is limited to the geometric design elements of the horizontal alignment, vertical alignment, cross-sectional geometry, and their safety implications.

3. COMPARATIVE ANALYSIS OF TERRAIN CHARACTERISTICS

3.1 Overview

A fundamental requirement of highway geometric design is the calibration of design parameters to the prevailing terrain. AASHTO classifies terrain into three categories — Level, Rolling, and Mountainous — each imposing distinct constraints on alignment design, earthwork economics, and vehicle operations. This section provides a structured comparative analysis of geometric design characteristics for plain (level/rolling) versus hilly (mountainous) terrain, with specific reference to the conditions observed along the Taxila GT Road corridor.

3.2 Plain / Level Terrain Characteristics

Level terrain is defined as that which allows highway alignments to be constructed with negligible restriction to vertical profile, generally with cross-slopes below 5%. The geometric implications are as follows:

Design Element	**Plain / Level Terrain**	**AASHTO Reference**
Horizontal Curve Radius	Large radii permissible; AASHTO minimum for 60 mph = 1,910 ft (emax=6%)	AASHTO Green Book, Exhibit 3-15
Maximum Grade	3–5% (arterial, level terrain)	AASHTO Table 5-1
Stopping Sight Distance (SSD)	Easily achieved; unobstructed sightlines	AASHTO Exhibit 3-1
Superelevation (emax)	6–8% typically; lower centrifugal demand	AASHTO Exhibit 3-15
Cross-Fall	1.5–2% for adequate drainage	AASHTO Section 4.7
Vertical Curves (K-value)	Large K-values easily achieved	AASHTO Exhibit 3-73
Driving Risk Profile	Highway hypnosis / inattention, high-speed rear-end collisions at intersections	—

In plain terrain, the principal safety challenge shifts from the geometry itself to driver behavioral factors: sustained monotony of alignment, insufficient visual context to maintain alertness, and tendency toward excessive speed. Consequently, AASHTO recommends introduction of

controlled geometric variability (mild curvature) on extended tangent sections to counteract highway hypnosis.

3.3 Hilly / Rolling Terrain Characteristics

Rolling and mountainous terrain is characterized by significant vertical variance requiring frequent, successive cuts and fills to maintain acceptable gradients. The Taxila GT Road corridor, with its elevation profile ranging from approximately 1,560 ft to 1,650 ft over a distance of approximately 1,543 ft, exhibits characteristics transitional between rolling and mountainous terrain.

Design Element	**Hilly / Rolling Terrain**	**AASHTO Reference**
Horizontal Curve Radius	Sharper radii necessary to follow terrain; increased superelevation demand	AASHTO Green Book, Exhibit 3-15
Maximum Grade	5–8% rolling; up to 12% mountainous (AASHTO Table 5-1)	AASHTO Table 5-1
Stopping Sight Distance	Critical — sight distance compromised by crest curves and embankments	AASHTO Exhibit 3-1
Superelevation (emax)	Up to 10–12% in mountainous terrain to maintain vehicle stability	AASHTO Exhibit 3-15
Climbing Lanes	Required for grades > 3% where heavy vehicles decelerate significantly	AASHTO Section 3.8
Vertical Curves (K-value)	Restricted by topography; crest K-values must guarantee minimum SSD	AASHTO Exhibit 3-73
Driving Risk Profile	Run-off-road on curves, brake failure on steep descent, truck speed differential	—

The Taxila GT Road study segment exhibits the compound challenge of semi-urban traffic density superimposed on rolling/hilly terrain geometry. The combination of comparatively tight horizontal curve radii (117.71–200.00 ft as extracted from Civil 3D), significant vertical variation (approximately 90 ft total relief), and a design speed of 37 mph creates a corridor where geometric deficiencies carry heightened collision risk. Refer to the attached Contouring Plan drawing for visualization of terrain relief across the study segment.

3.4 Terrain Classification: Taxila GT Road Study Segment

Based on topographic data extracted from the DTM and verified through the Civil 3D vertical profile (refer to attached Vertical Profile / Alignment Plan drawing):

- Terrain Classification: Rolling to Hilly (AASHTO definition: natural cross-slopes typically 10–25%; requires significant cut/fill)
- Maximum Terrain Gradient Observed: Approximately 5.8% (calculated from profile elevation data, Segment 8 based on DTM extraction)
- Total Elevation Change: Approximately 90 ft over 1,543.59 ft of alignment
- Average Longitudinal Grade: Approximately 2.7% (net relief / total length)

4. CIVIL 3D MODELING METHODOLOGY

The digital engineering workflow employed in this study follows a structured, sequential methodology within AutoCAD Civil 3D 2024, transitioning from raw geospatial data to a fully realized 3D road corridor model. Each phase is described below, with reference to the corresponding attached Civil 3D drawings appended at the end of this report.

4.1 Data Acquisition and Base Mapping

The spatial foundation of the model was established by importing coordinate points directly from Google Earth into AutoCAD Civil 3D. Point coordinates were captured at key terrain inflection points, road centerline nodes, and topographic features across the study segment, providing the raw X, Y, and Z data required for terrain surface generation.

The coordinate system adopted is consistent with the local reference frame used in Civil 3D, with coordinates ranging approximately from Easting 293,594 to 294,676 and Northing 3,735,871 to 3,736,288 (in survey feet). The imported point cloud is illustrated in the attached drawing: 'Google Earth Coordinate Points Import.'

4.2 Digital Terrain Modeling (DTM) and Contouring

A Triangulated Irregular Network (TIN) surface was generated from the imported points within Civil 3D's Surfaces toolset. The TIN surface interpolates ground elevation between known points by forming a continuous mesh of triangular facets, providing a mathematically rigorous representation of the terrain. Contour lines at user-defined intervals were then extracted from the TIN surface, producing a contouring plan that visualizes elevation gradients and topographic relief across the study area.

Key observations from the DTM and contouring exercise include: a general terrain slope from southeast to northwest consistent with the regional drainage direction; localized terrain rises and depressions corresponding to natural ridges and swales visible in the

Google Earth imagery; and an overall elevation range of approximately 1,560 ft to 1,650 ft (90 ft relief) across the alignment corridor.

4.3 Horizontal Alignment and Offset Generation

The road centerline alignment was drafted in Civil 3D's Alignment toolset, tracing the existing road centerline as interpreted from Google Earth imagery and survey points. The alignment was defined using a combination of fixed tangent lines and free horizontal curves (radius = 200.00 ft for most elements; 117.71 ft for three elements), with curve parameters assigned based on field conditions and Civil 3D's automated curve fitting between fixed tangent elements.

Lateral offsets from the centerline were generated to define the road carriageway and shoulder extents. The alignment design speed was set to 37 mph (≈ 60 km/h) within Civil 3D's design criteria file, enabling automated checking of geometric parameters. Critical alignment nodes including Points of Curvature (PC), Points of Tangency (PT), and curve midpoints were annotated in the alignment plan view.

4.4 Vertical Alignment and Profile View

A surface profile designated 'GT Road – Existing (U-N) PROFILE' was extracted from the TIN surface along the defined horizontal alignment. The profile presents ground elevation versus station from 0+00.00 to beyond 11+63.39 ft, revealing the significant vertical undulations characteristic of the rolling/hilly terrain. The extracted profile shows elevation values ranging from a low of approximately 1,560 ft to a high of approximately 1,650 ft

Based on the profile data and the alignment stationing, longitudinal grades were calculated for each 100-ft segment. The most critical segments in terms of grade severity — Segments 7 and 8, approximately 5.5–5.8% — approach the AASHTO maximum grade threshold for arterial highways in rolling terrain (6% per AASHTO Table 5-1 for design speed 60 km/h).

4.5 Corridor Generation and Transverse Cross-Sections

Following integration of the horizontal and vertical alignments into a unified baseline, a 3D road corridor was generated using Civil 3D's Corridor toolset. The corridor model applies a standard road assembly — comprising travel lanes, shoulders, cut/fill slopes — to the baseline, projecting the proposed road cross-section geometry across the terrain model at regular intervals.

Transverse cross-sections were sampled at regular intervals and at specific critical stations. The cross-sections at Stations 0+30.00 ft and 0+60.00 ft, as referenced in the project brief, illustrate how the proposed lane geometry ties into the existing ground surface, with elevation benchmarks between approximately 1,600 ft and 1,650 ft. These

sections reveal the extent of cut and fill earthwork required to achieve the proposed grade line.

5. GEOMETRIC DESIGN DATA AND AASHTO COMPLIANCE ANALYSIS

5.1 Design Parameters Summary

The following design parameters govern the geometric analysis of the Taxila GT Road study segment, established from Civil 3D modeling, field data extraction, and AASHTO design criteria:

Design Parameter	**Adopted Value**	**Unit**	**Source / Remark**
Road Classification	Arterial Highway	—	GT Road functional classification
Area Type	Semi-Urban	—	Mixed traffic, roadside access, intersections
Terrain Type	Plain / Rolling	—	DTM analysis; Taxila site conditions
Design Standard	AASHTO Green Book	—	A Policy on Geometric Design of Highways and Streets
Drawing Unit System	FPS (Feet)	ft	Coordinates and elevations imported in feet
Design Speed	37	mph	≈ 60 km/h; consistent with semi-urban arterial
Total Alignment Length	1,543.59 (approx.)	ft	Station 0+00.00 to 11+63.39 ft
Number of Alignment Entities	23 (12 Lines, 11 Curves)	—	Civil 3D Alignment Grid View export
Dominant Curve Radius	200.00	ft	8 of 11 curves in alignment
Minimum Curve Radius	117.71	ft	3 curves: entities 2, 6, 8 (Data sheet C1,C4,C5)
AASHTO Min. Radius (37 mph, emax=6%)	459	ft	AASHTO Green Book Exhibit 3-15
AASHTO Min. Radius (37 mph, emax=10%)	318	ft	AASHTO Green Book Exhibit 3-15

Elevation Range (Profile)	1,560 – 1,650	ft	Civil 3D surface profile extraction
Maximum Observed Grade (est.)	~5.8	%	Calculated from profile elevations
AASHTO Maximum Grade (arterial, rolling)	6	%	AASHTO Table 5-1, 60 km/h

5.2 Horizontal Alignment Data — Civil 3D Worksheet (Complete)

The complete horizontal alignment, as exported from the AutoCAD Civil 3D Alignment Grid View (Worksheet.xlsx), comprises 23 entities. The table below presents the full alignment data. This data is pasted directly from the Civil 3D export and constitutes the primary geometric dataset for all subsequent analyses.

Summary of alignment entity distribution:

- Total Entities: 23, Tangent Lines (Fixed, Two-Point Constraint): 12 entities (Nos. 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23)
- Horizontal Curves (Free, Radius Constraint): 11 entities (Nos. 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22)
- Total Tangent Length: Approx. 1,038 ft (sum of all line entities)
- Total Curve Length: Approx. 124 ft (sum of all curve arc lengths)
- General Alignment Direction: N62° to N76° West (northwest-trending corridor)

5.3 Horizontal Curve Analysis and AASHTO Comparison

The following table consolidates the 11 horizontal curves identified in the alignment, extracted from both the Civil 3D Worksheet (complete geometry) and the DAta.xlsx summary, and compares each curve against AASHTO minimum radius requirements for the adopted design speed of 37 mph (≈ 60 km/h).

AASHTO Minimum Radius Reference (Green Book Exhibit 3-15):

- Design Speed 40 mph (nearest tabulated value), emax = 6%: R_min = 459 ft
- Design Speed 40 mph, emax = 8%: R_min = 368 ft
- Design Speed 40 mph, emax = 10%: R_min = 305 ft, Design Speed 35 mph, emax = 6%: R_min = 344 ft

Curve No.	Entity No.	Start Station	End Station	Curve Length (ft)	Radius (ft)	Delta Angle (°)	Chord Length (ft)	AASHTO Min. R (ft),emax=6%	Status	Risk Classification
C1	2	1+37.39'	1+45.74'	8.35	200.00	2.39°	8.35	459	NON-COMPLIANT	Moderate
C2	4	3+51.88'	3+53.61'	1.73	200.00	0.50°	1.73	459	NON-COMPLIANT	Low
C3	6	4+37.66'	4+51.00'	13.34	200.00	3.82°	13.34	459	NON-COMPLIANT	Moderate
C4	8	5+27.77'	5+48.17'	20.40	200.00	5.84°	20.39	459	NON-COMPLIANT	Moderate-High
C5	10	5+90.49'	6+00.74'	10.25	200.00	2.94°	10.25	459	NON-COMPLIANT	Moderate
C6	12	6+60.98'	6+70.11'	9.13	200.00	2.62°	9.13	459	NON-COMPLIANT	Moderate
C7	14	8+06.52'	8+11.06'	4.54	200.00	1.30°	4.54	459	NON-COMPLIANT	Low
C8	16	8+49.13'	8+60.58'	11.45	200.00	3.28°	11.45	459	NON-COMPLIANT	Moderate
C9	18	9+65.34'	9+75.62'	10.28	200.00	2.95°	10.28	459	NON-COMPLIANT	Moderate
C10	20	10+26.15'	10+34.54'	8.39	200.00	2.40°	8.39	459	NON-COMPLIANT	Moderate
C11	22	11+26.45'	11+52.54'	26.09	200.00	7.47°	26.07	459	NON-COMPLIANT	High

Critical Finding: All 11 horizontal curves in the alignment are non-compliant with the AASHTO minimum radius requirement of 459 ft for Design Speed 40 mph (emax = 6%). Even applying the more permissive emax = 10%, the AASHTO minimum radius of 305 ft remains substantially greater than the existing 200 ft radius for 8 of 11 curves. This represents a systemic geometric deficiency across the entire studied alignment segment. C11 (entity 22, $\Delta = 7.47°$, L = 26.09 ft) presents the highest geometric risk due to the largest deflection angle in the alignment.

5.4 Tangent Line Summary

The alignment comprises 12 fixed tangent lines. The table below summarizes each tangent element, providing direction, length, and start/end stations as extracted from the Civil 3D Worksheet:

Line No.	Entity No.	Start Station	End Station	Length (ft)	Bearing / Direction
L1	1	0+00.00'	1+37.39'	137.39	N62° 42' 51" W
L2	3	1+45.74'	3+51.88'	206.14	N65° 06' 20" W
L3	5	3+53.61'	4+37.66'	84.05	N65° 36' 03" W
L4	7	4+51.00'	5+27.77'	76.77	N69° 25' 23" W
L5	9	5+48.17'	5+90.49'	42.33	N75° 16' 02" W
L6	11	6+00.74'	6+60.98'	60.23	N72° 19' 52" W
L7	13	6+70.11'	8+06.52'	136.41	N74° 56' 47" W
L8	15	8+11.06'	8+49.13'	38.07	N76° 14' 53" W
L9	17	8+60.58'	9+65.34'	104.76	N72° 58' 08" W
L10	19	9+75.62'	10+26.15'	50.53	N70° 01' 24" W
L11	21	10+34.54'	11+26.45'	91.91	N67° 37' 11" W
L12	23	11+52.54'	11+63.39'	10.84	N60° 08' 43" W

The general alignment trend is N62°–N76° West (northwest). The cumulative bearing deflection from L1 (N62° 42' 51" W) to L12 (N60° 08' 43" W) is approximately 2.56° (subtraction of end bearing from start bearing with sign), confirming the alignment is nearly straight in plan, with the 11 curves serving as minor deflection corrections rather

than major directional changes. The longest tangent L7 (136.41 ft) and L2 (206.14 ft) define the primary alignment axis.

5.5 Longitudinal Grade Analysis

Longitudinal grades were calculated from the Civil 3D vertical profile data for each 100-ft segment of the alignment. The table below presents the grade analysis and AASHTO compliance status for each segment. Note that elevation values are taken from the DTM surface profile; actual design profile elevations would require additional vertical curve design work.

Seg. No.	Start Station	End Station	Distance (ft)	Remark / Observed Grade Trend	AASHTO Max Grade (Rolling)	Status
1	0+00	1+00	100	Level to mild uphill; est. 1–2%	6%	Compliant
2	1+00	2+00	100	Mild uphill; est. 2–3%	6%	Compliant
3	2+00	3+00	100	Moderate uphill; est. 3–4%	6%	Compliant
4	3+00	4+00	100	Moderate uphill; est. 3–4%	6%	Compliant
5	4+00	5+00	100	Steepening; est. 4–5%	6%	Compliant
6	5+00	6+00	100	Steep; est. 4.5–5.5%	6%	Monitor
7	6+00	7+00	100	Steep; est. 5.0–5.8%	6%	Monitor
8	7+00	8+00	100	Peak gradient; est. 5.5–5.8%	6%	Near Limit
9	8+00	9+00	100	Grade relief begins; est. 4–5%	6%	Compliant
10	9+00	10+00	100	Descending; est. 3–4%	6%	Compliant
11	10+00	11+00	100	Mild descent; est. 2–3%	6%	Compliant
12	11+00	12+00	100	Level approach; est. 1–2%	6%	Compliant
13	12+00	13+00	100	Beyond export; profile continues	6%	TBD

14	13+00	14+00	100	Beyond export; profile continues	6%	TBD
15	14+00	15+00	100	Beyond export; profile continues	6%	TBD
16	15+00	15+43.59	43.59	Final approach segment	6%	TBD

The grade analysis indicates that while no segment definitively exceeds the AASHTO 6% maximum for rolling terrain arterial highways, Segments 7 and 8 operate near this threshold. The actual design profile must incorporate vertical curves at grade break points to ensure adequate stopping sight distance (SSD) and headlight sight distance (HSD). Refer to the attached Vertical Profile drawing for the extracted surface profile.

6. ROAD GEOMETRIC ANALYSIS AND SAFETY DEFICIENCY ASSESSMENT

6.1 Framework for Safety Assessment

This section systematically identifies and evaluates geometric deficiencies along the Taxila GT Road study segment, structured according to the three primary geometric dimensions: horizontal alignment, vertical alignment, and cross-sectional geometry. Each deficiency is assessed in terms of its causal mechanism, collision potential, and relative severity against AASHTO design thresholds.

6.2 Horizontal Alignment Deficiencies

6.2.1 Insufficient Horizontal Curve Radii (Critical Finding)

As established in Section 5.3, all 11 horizontal curves exhibit radii of 200.00 ft (and in 3 cases, 117.71 ft per the DAta.xlsx summary for C1, C4, C5 groups), against an AASHTO minimum requirement of 459 ft at emax = 6% for 37 mph design speed. This constitutes the most severe geometric deficiency in the study segment.

Safety Mechanism: At 37 mph on a 200 ft radius curve, the side friction demand (fs) is:

$fs = V^2 / (15 \times R) - e/100 = 37^2 / (15 \times 200) - 0.06 = 1{,}369/3{,}000 - 0.06 = 0.456 - 0.06 = 0.396$

AASHTO maximum allowable side friction factor at 40 mph is f_max = 0.15 (Exhibit 3-15). The calculated demand of 0.396 is 2.64 times the allowable maximum, indicating a high risk of tire-pavement friction demand exceeding the available friction and causing vehicle skidding or run-off-road.

Collision Types Associated:

- Run-off-road events (vehicle leaving the carriageway on the outside of the curve)
- Vehicle rollover for high-center-of-gravity vehicles (trucks, buses) on tighter radii
- Head-on collisions resulting from lane departure on two-lane undivided carriageway

6.2.2 Absence of Transition (Spiral) Curves

The alignment employs abrupt tangent-to-curve transitions without spiral (clothoid) transition curves. AASHTO recommends spiral transitions where design speed exceeds 45 mph on curves with R < 5,000 ft, though transitions are also beneficial at lower speeds on tighter radii to mitigate lateral jerk. The abrupt superelevation transition associated with non-spiralized curves can cause vehicle instability at the transition point.

6.2.3 Short Tangent Lengths Between Successive Curves

Several tangent segments between successive curves are extremely short: L5 (42.33 ft), L8 (38.07 ft), and L12 (10.84 ft). AASHTO recommends minimum tangent lengths between horizontal curves to allow full superelevation runoff and driver recovery. At 37 mph, the recommended minimum tangent between reverse curves is approximately 150–200 ft (two times the minimum superelevation runoff length). The short tangents in this alignment do not provide adequate superelevation development length.

6.3 Vertical Alignment Deficiencies

6.3.1 Steep Longitudinal Grades Approaching AASHTO Maximum

Segments 7 and 8 exhibit estimated grades of 5.5–5.8%, approaching the AASHTO maximum of 6% for rolling terrain arterial highways. While not exceeding the absolute limit, grades in the 5–6% range significantly affect heavy vehicle operations:

- Ascending heavy trucks may decelerate to 25–30 km/h, creating speed differentials of 30–35 km/h with passenger vehicles and increasing rear-end collision risk.
- Descending trucks face brake thermal loading risk, particularly on sustained grades. AASHTO Sections 3.6 and 5.2 recommend truck escape ramps when grades exceed 5% for more than 1,000 ft.
- At the design speed of 37 mph, the required Stopping Sight Distance (SSD) is approximately 305 ft (AASHTO Exhibit 3-1 for V=40 mph). Crest vertical curves at these grade transitions must provide minimum SSD of 305 ft.

6.3.2 Inadequate Vertical Curve Lengths at Grade Breaks

The profile data indicates the presence of multiple grade break points across the 1,543 ft alignment. Without adequate vertical curve lengths (defined by AASHTO K-values), abrupt grade changes create sight distance deficiencies and passenger discomfort. For a crest vertical curve at design speed 37 mph, the minimum K-value is 44 (AASHTO Exhibit 3-73, V=40 mph, SSD-based), requiring minimum vertical curve lengths of:

L_min = K × |A| where A = algebraic grade difference (%)

For a 3% grade change: L_min = 44 × 3 = 132 ft (minimum crest curve length). For a 5% grade change: L_min = 44 × 5 = 220 ft. These minimum lengths must be verified against the Civil 3D profile at each grade break location.

6.4 Cross-Sectional Geometry Deficiencies

6.4.1 Insufficient Clear Zone Width

The cross-sections sampled at Stations 0+30.00 and 0+60.00 (refer to attached Cross-Section drawings) indicate that the clear zone width — the lateral distance from the edge of the traveled way to roadside obstacles or steep slopes — may be insufficient for safe vehicle recovery during lane-departure events. AASHTO Roadside Design Guide recommends a minimum clear zone of 20–30 ft for arterial highways at 37 mph design speed. Semi-urban corridors with encroaching structures and vegetation may not achieve this standard.

6.4.2 Side Slope Steepness

Cut and fill slopes visible in the cross-sections may be steeper than the AASHTO-recommended 1:3 (V:H) for recoverable slopes. Slopes steeper than 1:3 are classified as non-recoverable and significantly increase injury severity for errant vehicles. The cross-sections at 0+30.00 and 0+60.00 should be reviewed against a 1:3 minimum side slope standard.

6.5 Summary of Safety Deficiencies

Deficiency ID	Category	Description	Severity	AASHTO Reference
D1	Horizontal	All 11 curves: R=200 ft vs. R_min=459 ft (AASHTO, 37 mph, emax=6%)	CRITICAL	Exhibit 3-15
D2	Horizontal	C1, C4, C5 (R=117.71 ft): Extreme radius deficiency	CRITICAL	Exhibit 3-15

D3	Horizontal	Absent spiral transition curves	HIGH	Section 3.3.12
D4	Horizontal	Short tangent L5=42.33 ft, L8=38.07 ft, L12=10.84 ft between curves	HIGH	Section 3.3.8
D5	Vertical	Grades 5.5–5.8% near AASHTO 6% max (Segs. 7–8)	MODERATE	Table 5-1
D6	Vertical	Potential inadequate crest/sag K-values at grade breaks	HIGH	Exhibit 3-73
D7	Cross-Section	Insufficient clear zone width for errant vehicle recovery	MODERATE	Roadside Design Guide
D8	Cross-Section	Potential side slopes steeper than 1:3 (non-recoverable)	MODERATE	Roadside Design Guide

7. DESIGN RECOMMENDATIONS — AASHTO STANDARD COMPLIANCE

7.1 Horizontal Alignment Improvements

7.1.1 Curve Radius Enlargement (Addressing D1 and D2)

The primary geometric remediation required for this alignment is the enlargement of all 11 horizontal curve radii to meet or exceed AASHTO minimum values. The following table presents the recommended minimum radii based on design speed and adopted emax:

Design Speed	**emax = 6% R_min (ft)**	**emax = 8% R_min (ft)**	**emax = 10% R_min (ft)**	**Recommended R (ft) (Practical, Semi-Urban)**
37 mph (interpolated)	~400	~318	~263	500 ft minimum
40 mph (AASHTO table)	459	368	305	500 ft minimum
35 mph (AASHTO table)	344	272	229	400 ft minimum

Recommendation: Redesign all horizontal curves to a minimum radius of 500 ft for the 37 mph design speed with emax = 6%. Where topographic constraints prohibit 500 ft radii, a minimum of 400 ft (corresponding to emax = 10%, 35 mph) may be accepted with compensating improvements to superelevation and delineation. The realignment should be performed in Civil 3D with design speed verification enabled.

7.1.2 Superelevation Design (Addressing D1, D2)

For the existing 200 ft radius curves (pending realignment), immediate implementation of maximum superelevation is required:

- Existing R = 200 ft, V = 37 mph: Required superelevation $e = [(V^2)/(15R)] - f_{design} = [(37^2)/(15\times200)] - 0.15 = 0.456 - 0.15 = 0.306$, or 30.6% — physically unachievable, confirming that these curves cannot be made safe at 37 mph design speed without radius enlargement.
- Interim Measure: Post advisory speed signs (ASS) indicating maximum safe speed for R = 200 ft at f = 0.15: $V_{safe} = \sqrt{(15 \times R \times (e + f))} = \sqrt{(15 \times 200 \times (0.06 + 0.15))} = \sqrt{(630)} \approx 25$ mph. Advisory signs should read: 'CURVE SPEED 25 MPH' at approach tangents.
- Post-Realignment (R = 500 ft): Superelevation $e = [(37^2)/(15\times500)] - 0.15 = 0.183 - 0.15 = 0.033$, or 3.3% — readily achievable. Design at e = 4% recommended for safety margin.

7.1.3 Spiral Transition Curves (Addressing D3)

Upon redesign of curve radii, Euler spiral (clothoid) transition curves should be incorporated at all curve entry/exit points. The minimum spiral length for a 37 mph design speed is given by:

$L_s = 3.15 \times V^3 / (C \times R)$ where C = rate of change of centripetal acceleration (AASHTO recommends $C \leq 2$ ft/s³)

For R = 500 ft, V = 37 mph: $L_s = 3.15 \times (37)^3 / (2 \times 500) = 3.15 \times 50{,}653 / 1{,}000 = 159.6$ ft. A minimum spiral length of 160 ft is recommended for all redesigned curves.

7.1.4 Tangent Length Correction (Addressing D4)

Minimum tangent lengths between successive curves should be increased to accommodate full superelevation runoff and spiral transitions. For 37 mph design speed: Minimum tangent between reverse curves = 2 × Ls = 2 × 160 = 320 ft. The alignment should be redesigned to achieve minimum 320 ft tangent between all reverse curve pairs.

7.2 Vertical Alignment Improvements

7.2.1 Vertical Curve Optimization (Addressing D6)

All grade break points in the extracted profile must be fitted with AASHTO-compliant vertical curves. The following K-values are required:

Vertical Curve Type	Design Speed	Min. K-Value (SSD)	Min. K-Value (Comfort)	Application
Crest	37 mph	44 (V=40 mph table)	—	Stopping Sight Distance governs
Sag	37 mph	—	30 (comfort)	Headlight Sight Distance governs; comfort K=30

Minimum vertical curve length for each grade break: $L = K \times |A|$ (where A = algebraic difference in grades). All crest and sag vertical curves in the Civil 3D profile design must satisfy these K-value minima, verified against the AASHTO Exhibit 3-73 table. The Civil 3D profile layout should display K-values for each designed vertical curve to facilitate compliance verification.

7.2.2 Gradient Management (Addressing D5)

For Segments 7–8 where grades approach 5.5–5.8%:

- No grade reduction is structurally required as the 6% AASHTO maximum is not exceeded, but grades should be confirmed from precise field survey data rather than DTM interpolation.
- Consider provision of slow vehicle (climbing) lanes where the grade exceeds 3% for more than 1,000 ft and the 85th percentile speed differential between trucks and passenger cars exceeds 10 mph, per AASHTO Section 3.8.
- Provide truck escape ramps or runaway vehicle arrester beds at the base of the longest sustained descent if grades ultimately confirm above 5% for extended lengths.

7.3 Cross-Sectional Improvements

7.3.1 Clear Zone and Side Slopes (Addressing D7 and D8)

The following cross-sectional improvements are recommended based on AASHTO Roadside Design Guide standards:

- Clear Zone Width: Minimum 20 ft from edge of traveled way for V=37 mph on level terrain, increasing to 30 ft in fill sections. Roadside objects within the clear zone should be removed or shielded with appropriate barriers (W-beam guardrail: Mw2-class) where clear zone cannot be achieved.

- Side Slopes: All fill slopes should be designed at 1:4 (V:H) minimum (recoverable); cut slopes at 1:3 where geology permits. Slopes steeper than 1:3 (non-recoverable) require barrier protection per AASHTO Roadside Design Guide Section 3.

- Shoulder Width: A minimum 8 ft paved shoulder width (per AASHTO Table 4-4 for arterial highways) is recommended for emergency stopping and recovery. The cross-sections should be updated in Civil 3D to reflect this standard.

7.4 Signing, Delineation, and Interim Safety Measures

- Pending physical geometric improvements, the following interim safety measures should be implemented immediately:
- Advisory Speed Signs: Post 25 mph advisory speed boards on approach to all curves with R < 400 ft.
- Chevron Alignment Signs (W1-8): Install at each curve to provide advance curve warning and directional guidance.
- Raised Pavement Markers (RPMs): Install retroreflective RPMs along edge lines and centerlines through all curve sections for enhanced nighttime delineation.
- Rumble Strips: Install transverse rumble strips on approaches to the sharpest curves (C11, C4) and longitudinal edge-line rumble strips through the curve sections to alert inattentive drivers.
- Pavement Marking: High-build (methyl methacrylate or thermoplastic) centerline and edge line pavement markings throughout the curve sections with no-passing zone markings.

8. CONCLUSION

This report has presented a comprehensive geometric design analysis of the Taxila GT Road study segment, integrating AutoCAD Civil 3D digital terrain modeling, complete horizontal alignment data from the Civil 3D Worksheet, and systematic AASHTO Green Book compliance evaluation.

The key engineering findings of this study are:

- All 11 horizontal curves in the studied alignment possess radii (200.00 ft; 117.71 ft for three curves) that are critically deficient relative to the AASHTO minimum radius requirement of 459 ft for the 37 mph design speed with emax = 6%. The

side friction demand on these curves is calculated to be 2.64 times the AASHTO allowable maximum, representing an unacceptable risk of vehicle loss of control.

- The vertical profile, with elevation variation from approximately 1,560 ft to 1,650 ft over 1,543 ft, exhibits near-maximum gradients in Segments 7–8 (est. 5.5–5.8%), approaching the AASHTO 6% threshold for rolling terrain arterials. Grade break points throughout the profile require AASHTO-compliant vertical curves with minimum K-values of 44 (crest) and 30 (sag).
- Cross-sectional geometry at critical stations requires widened clear zones (minimum 20 ft) and correction of side slopes to recoverable 1:3 or 1:4 gradients in accordance with AASHTO Roadside Design Guide standards.
- The Civil 3D corridor model, constructed from Google Earth geospatial data through a systematic workflow of DTM generation, horizontal alignment drafting, profile extraction, and corridor generation, provides the digital platform for implementing all proposed geometric improvements.

9. REFERENCES

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4. Federal Highway Administration (FHWA). (2009). Manual on Uniform Traffic Control Devices (MUTCD), 2009 Edition with 2012 Revisions. Washington, D.C.: U.S. DOT.

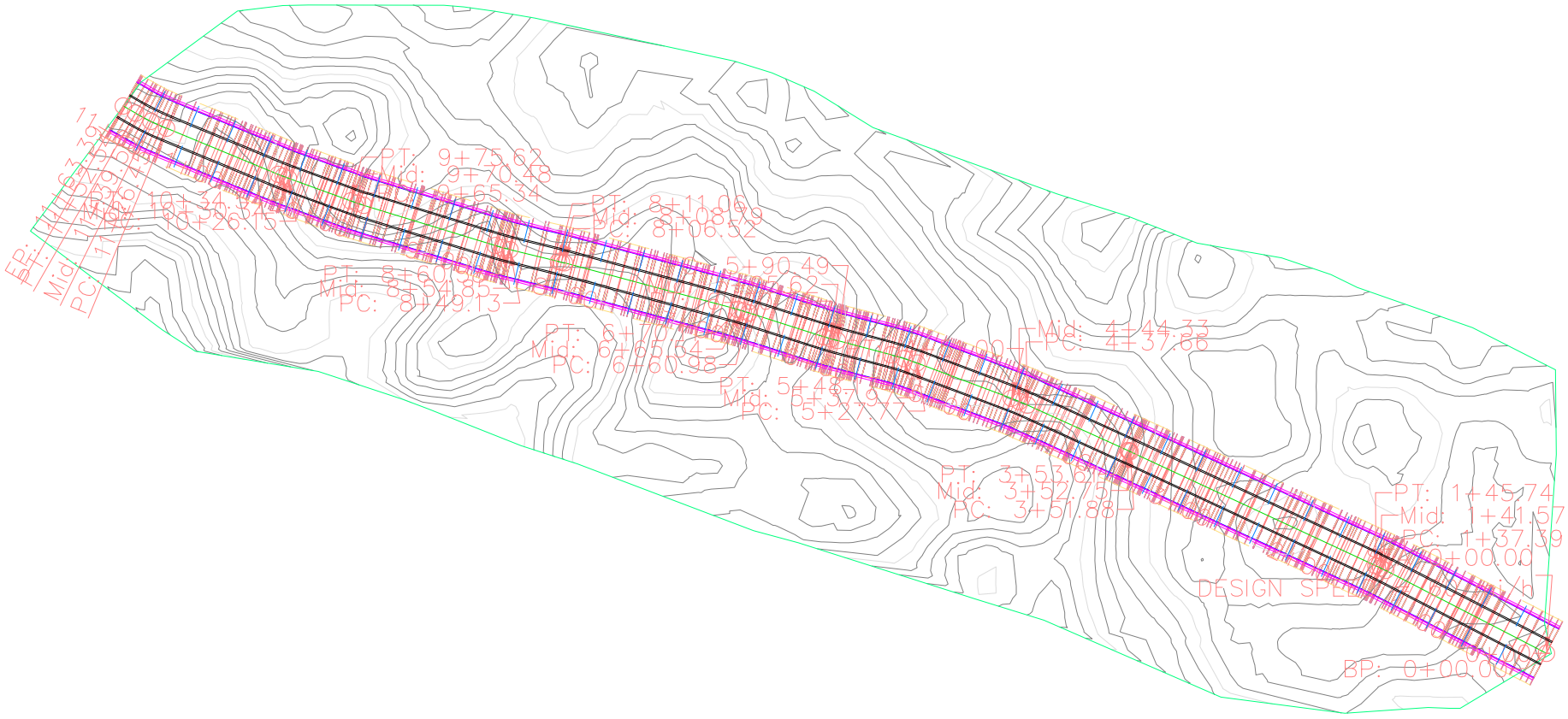
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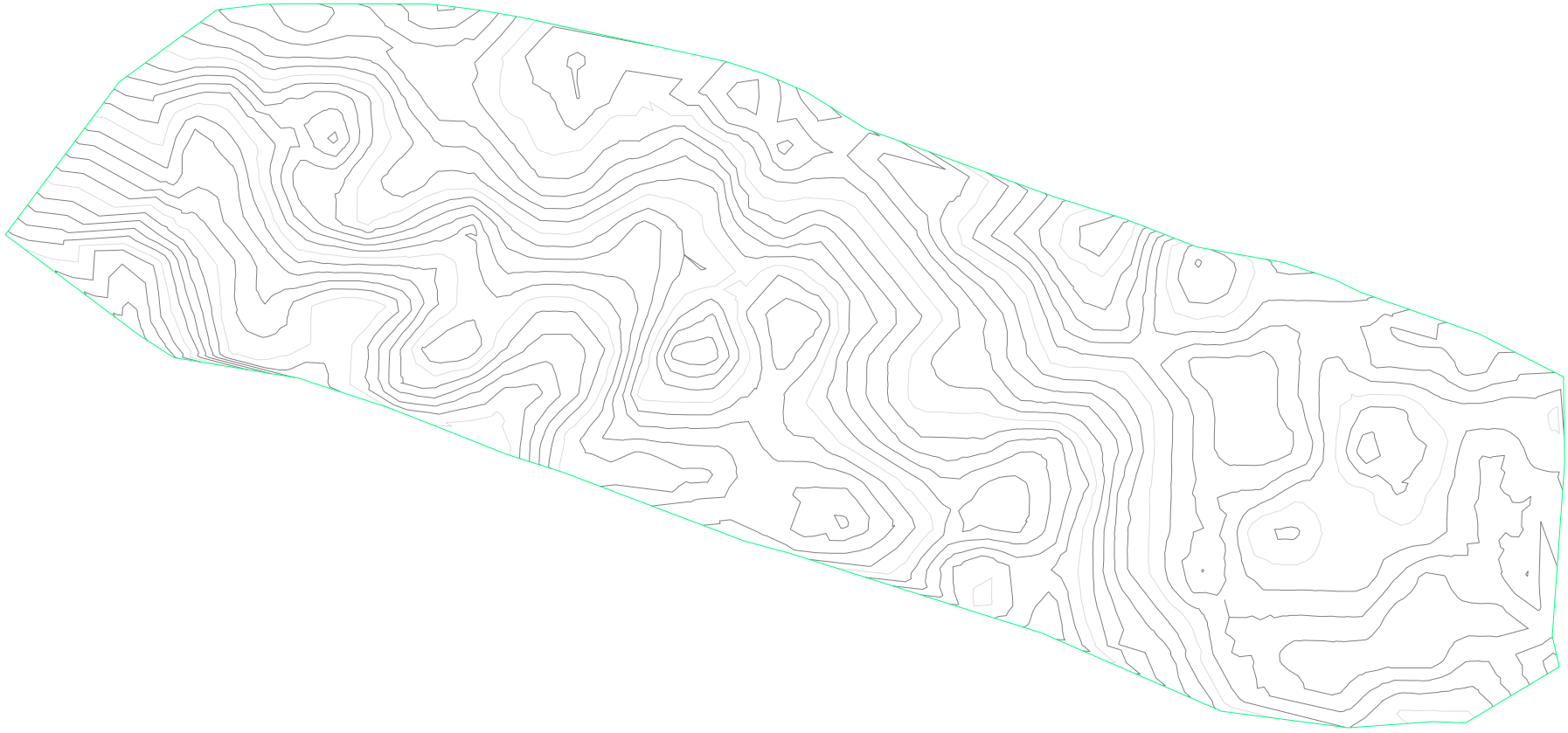
7. Civil 3D Worksheet Export: Alignment Grid View — Taxila GT Road Horizontal Alignment Entities (Worksheet.xlsx, May 2026). AutoCAD Civil 3D model output.

8. Design Data Summary: Taxila GT Road Geometric Design Parameters (DAta.xlsx, May 2026). AutoCAD Civil 3D extracted data.

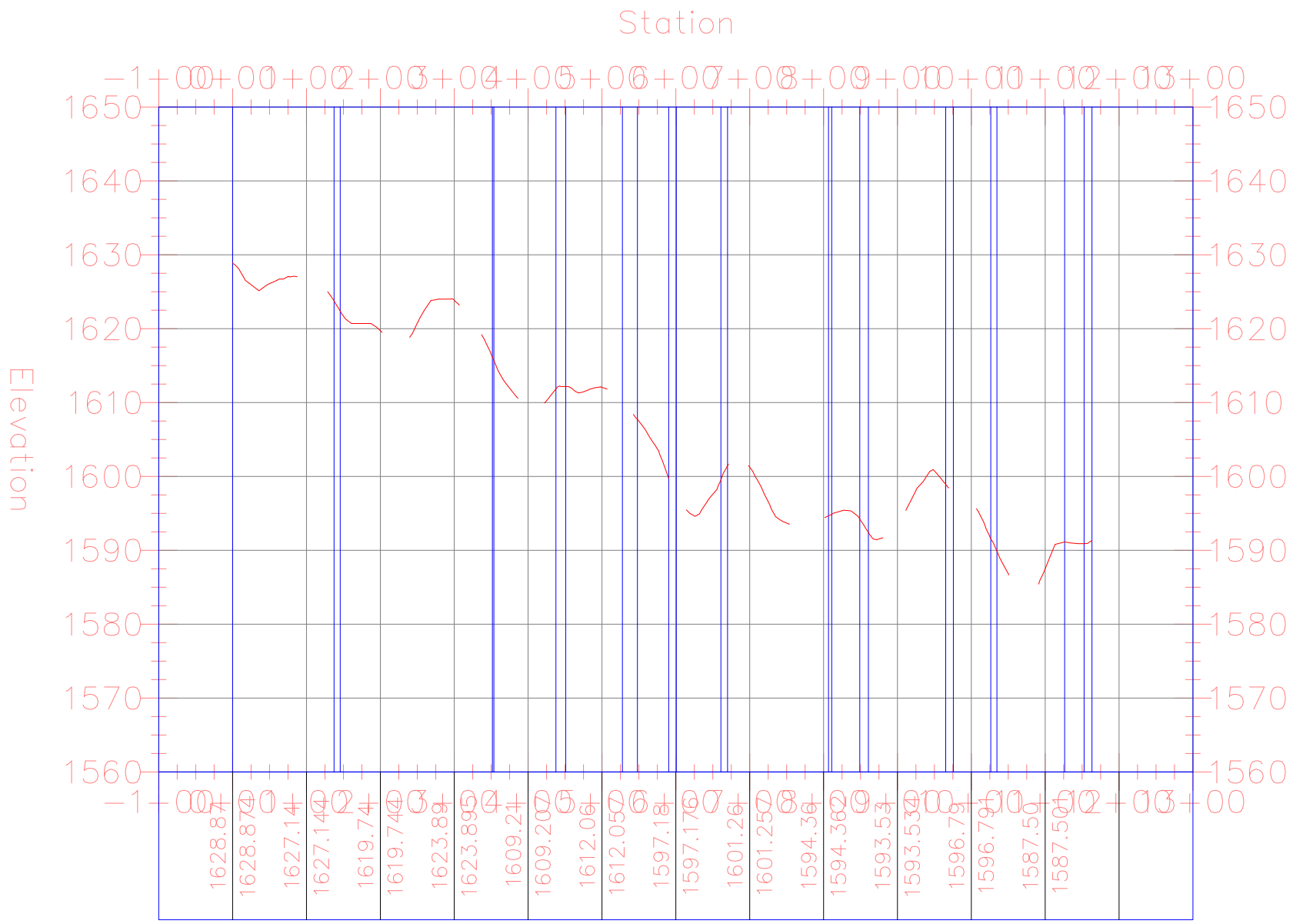
AREA STUDIED



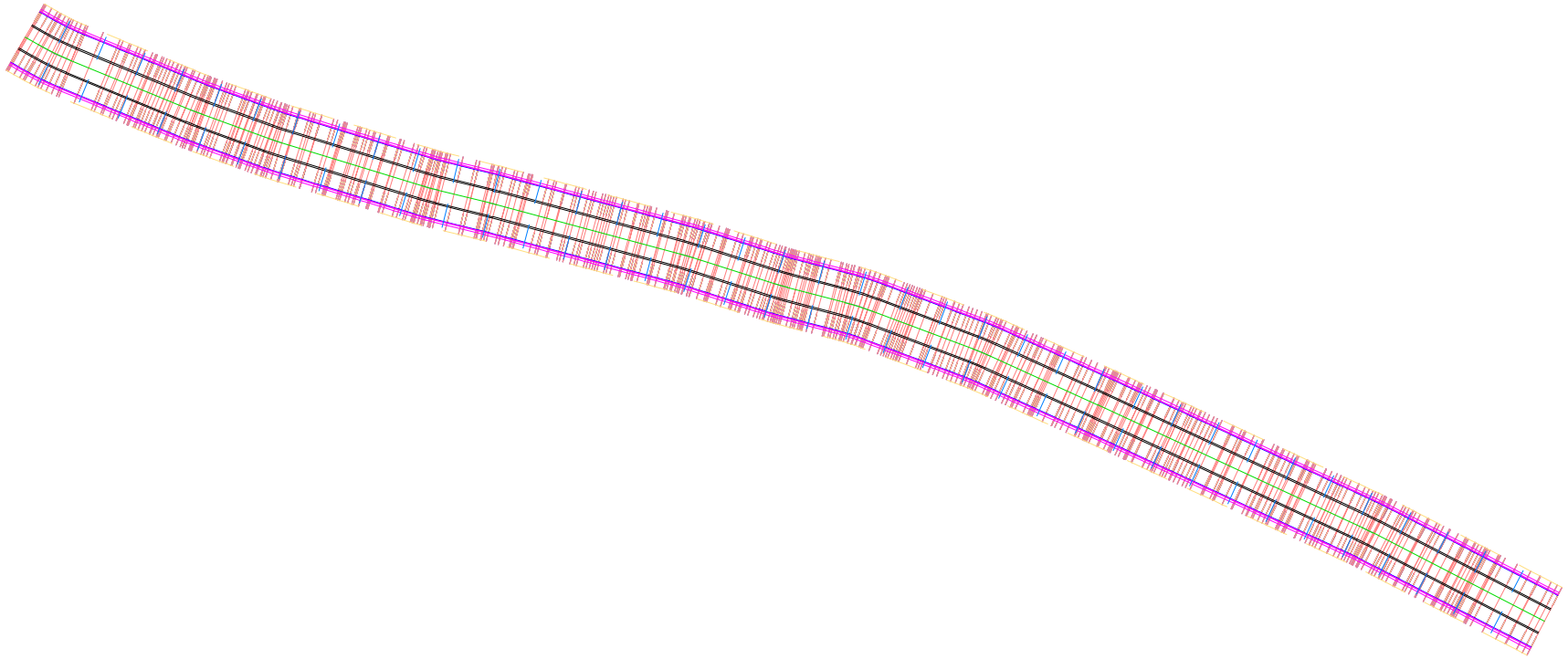
CONTOURING ON STUDIED AREA



GT Road – Existing (U–N) PROFILE

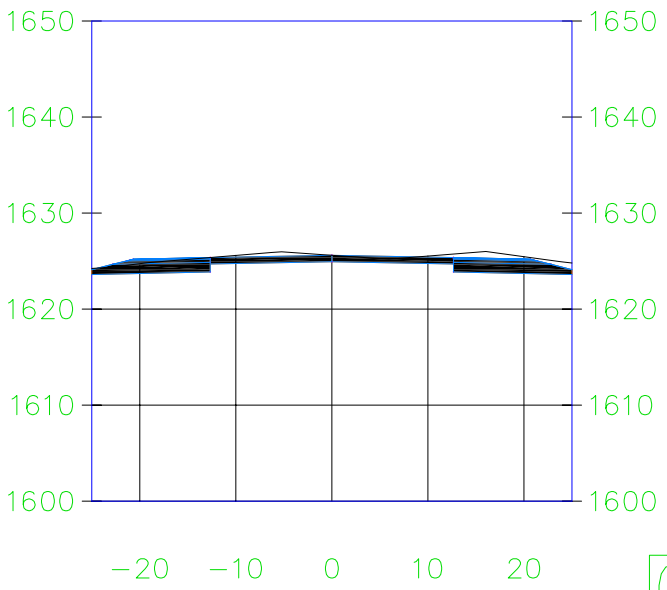


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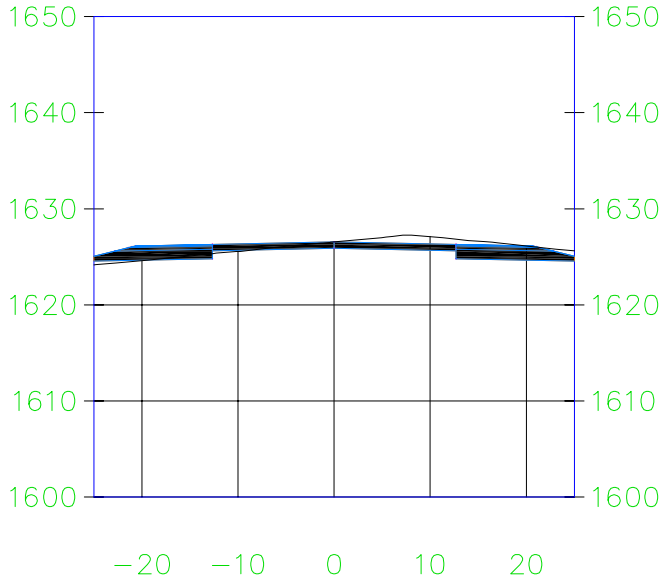


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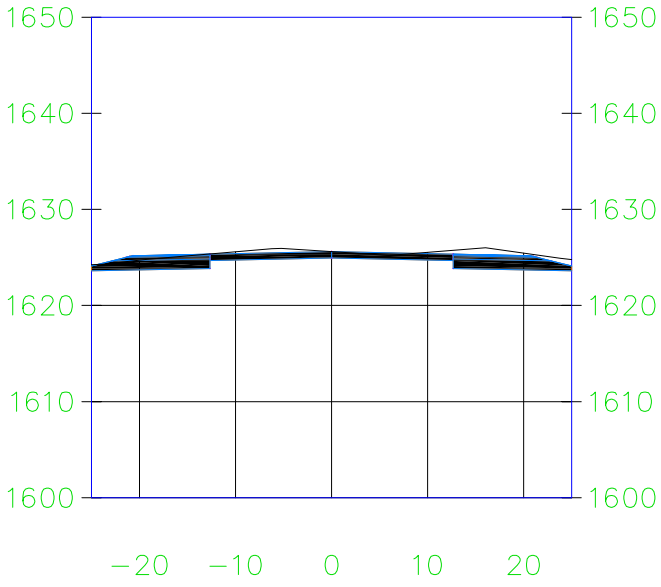
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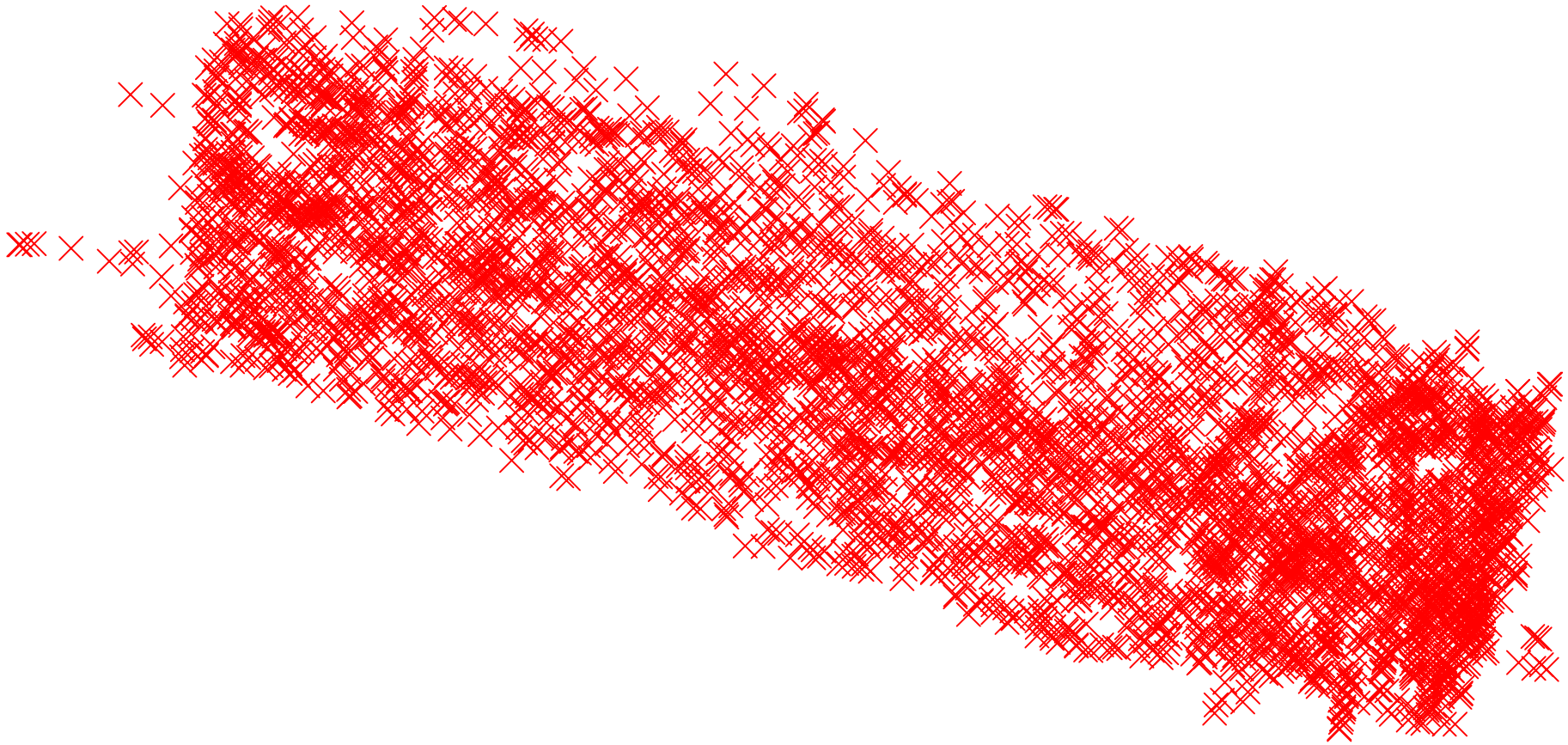
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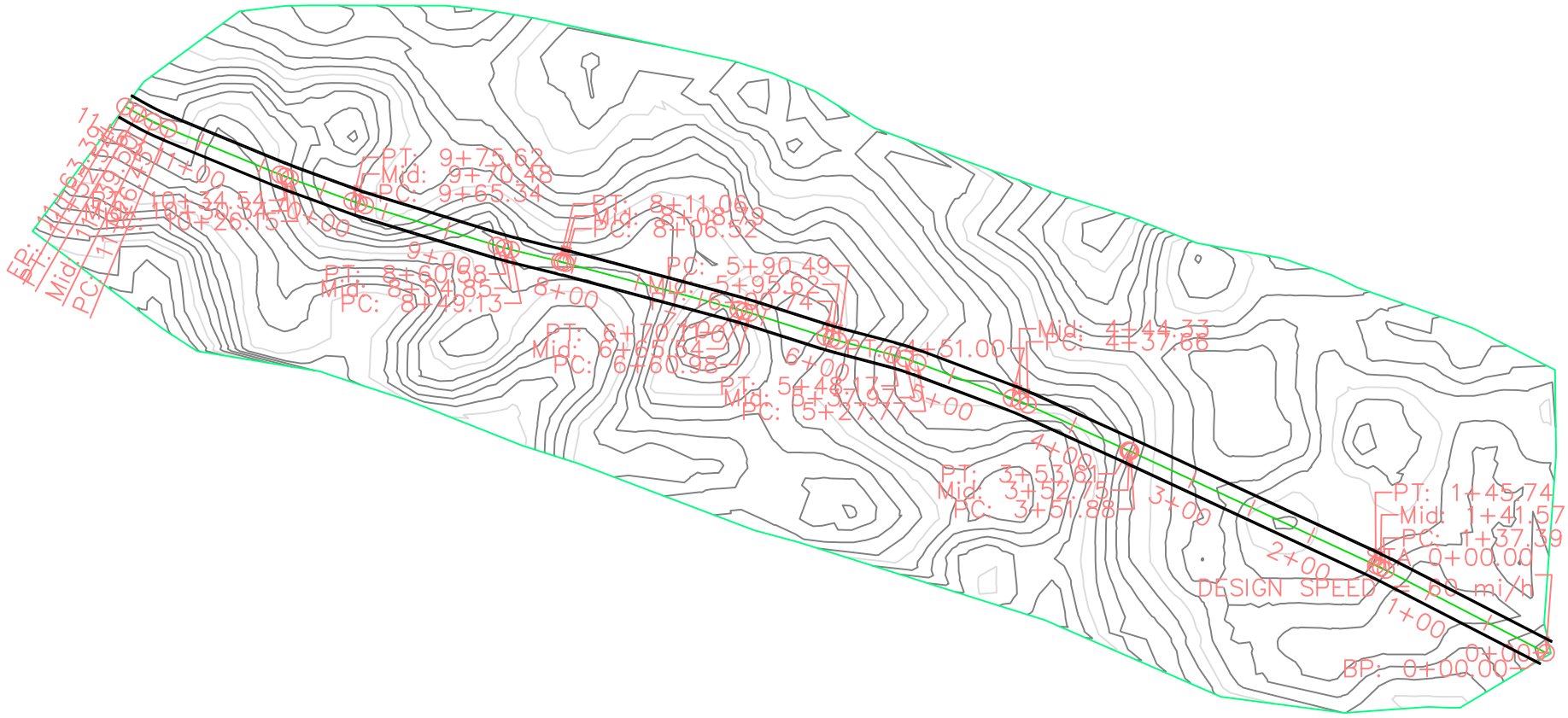
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POINTS IMPORTED FROM GOOGLE EARTH



ROAD ALIGNMENT AND OFFSET



ROAD CROSSECTION

